

A Work-in-Progress Study on Effect of Visual Pointer Speed on Perceived Pleasantness during Membrane Penetration with Force Feedback

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ABSTRACT

To facilitate haptic technology adoption, designing stimuli eliciting pleasant feelings is considered one important approach. However, how pleasantness can be effectively induced is not fully understood. We explore whether pleasantness is postdictively modulated by visual pointer speed. Participants manipulated the pointer to press and penetrate a membrane under different pointer speeds (press/release) and stiffness levels. Results suggest that speed effect may vary depending on the stiffness level. Pleasantness tended to be higher with moderately high release speeds under high stiffness, whereas slower pressing speeds tended to be preferred under no-stiffness conditions. These findings provide insights for designing pleasant haptic interactions.

Index Terms: Haptic, virtual Reality, force feedback, pleasantness, affective computing.

1 INTRODUCTION

Haptic presentation technologies, which stimulate somatic receptors, are rapidly advancing in both research and application contexts [6]. Various methods, ranging from force and vibration to electrical and thermal stimulation, have been proposed to artificially synthesize tactile feelings [3]. Such technologies can convey not only symbolic information to users but also physical properties of objects, such as weight and hardness [11], with promising applications extending from mobile gaming to high-fidelity surgical simulation in AR/VR [5, 1].

However, for haptic presentation systems to achieve widespread adoption, they must go beyond merely rendering physical attributes. They would also be better to be designed to elicit pleasantness. Given the barriers of cost, size, and complexity often associated with haptic hardware [11], the capacity to evoke positive affect, or "hedonic motivation," becomes a critical factor in driving user adoption and sustained use [10]. If a haptic presentation systems can induce pleasantness during interaction, it may significantly enhance the user's overall impression and thereby offset technical limitations.

Previous research on eliciting pleasantness through active touch has mainly focused on texture exploration of object surfaces, and little is known about interactions involving dynamic deformation or penetration of objects. Pleasantness has been shown to be associated with specific textural properties such as friction [7]. Also, the links between physical and emotional descriptors across diverse materials have been examined [4]. However, the pleasantness experienced during manipulations that involve large deformation or penetration remains largely underexplored.

This work-in-progress study focuses on the pleasantness experienced when pressing and penetrating a virtual membrane with a pointer while receiving resistance. Among various prototype

demonstrations we explored, this task was identified as one of the most pleasant. Thus, we adopted it as a case study. To investigate whether pleasantness is influenced by visual pointer speed, we manipulated its speed during pressing and after penetration. We also varied the resistance force experienced during the pressing phase. Although the current report is based on preliminary experiments with a limited number of participants, we aim to extend this work to further explore how pleasantness can be induced, and to discuss implications for the design of haptic interactions that aim to evoke pleasantness.

2 PRELIMINARY EXPERIMENT

In this experiment, we investigated whether pleasantness is modulated postdictively. A total of six Japanese participants took part in the experiment. All participants self-reported being right-handed. Ethical approval for this study was obtained from the Ethics Committee of the University of Tokyo (Approval number: 24-583).

2.1 Stimuli

In this experiment, we employed a task in which participants manipulated a pointer to push and penetrate a membrane (see Figure 1A).

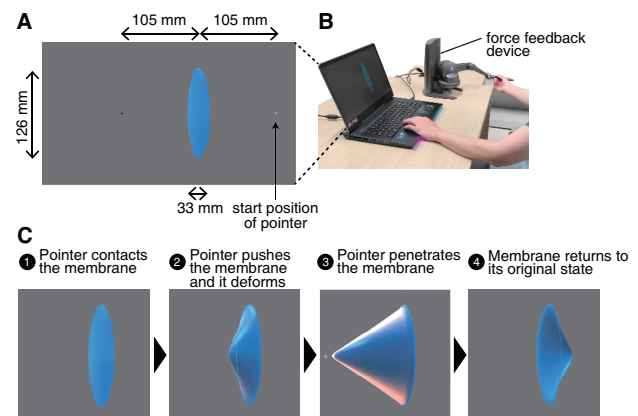


Figure 1: Experimental environment. A, Experimental window. B, Apparatus. C, The deformation of the membrane.

Figure 1B shows the apparatus. A laptop PC (MSI GE66 Raider 11UG, Intel Core i7-11800H, 2.3 GHz) was placed on a desk, and participants were seated approximately 60 cm away from the display. They held the end effector of a force feedback device (TouchX, 3D Systems Corporation) using their thumb, index, and middle fingers. To eliminate ambient noise, participants wore noise-canceling headphones (WH-1000XM5, Sony).

The interaction scenario was visualized and presented on a computer screen. During each trial, participants used the end effector of the force feedback device to manipulate the on-screen pointer. They were instructed to push the membrane to the left using the pointer (see Figure 1C). The membrane deformed as it was pushed by the pointer. After the membrane was deformed to its maximum

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allowable displacement (105 mm), the pointer penetrated through the membrane. Following the penetration, the membrane returned to its original state.

We manipulated the visual pointer speeds during the pressing phase and after penetration (i.e., press speed and release speed). The press speed was manipulated at three levels (ratios of 0.25, 0.5, and 1.0). A ratio of 1.0 corresponded to the default pointer speed, while 0.5 indicated half that speed. The release speed was manipulated at five levels (ratios of 0.25, 0.5, 1.0, 2.0, and 4.0). Additionally, the membrane's stiffness, which corresponded to the resistance force physically presented by the TouchX device during membrane deformation, was varied across three levels (0, 0.2, and 0.4 N/mm). In the 0 N/mm condition, no force was presented. At the moment of penetration, the force presented by the device was immediately set to zero. In total, this resulted in 45 experimental conditions. The membrane deformation during pressing was simulated using Obi Softbody 7 in Unity¹. After penetration, the membrane deformed to oscillate in simple harmonic motion around its origin.

2.2 Procedure

In each trial, participants pushed and penetrated the membrane using a pointer. No specific instructions were given regarding the speed of the pointer movement. Five seconds after the pointer had penetrated the membrane, a response screen was displayed.

On the response screen, participants saw the prompt: "Please judge the pleasantness." Responses were collected using a Visual Analog Scale (VAS) [2]. Participants were instructed to click farther to the right on the scale if they perceived the experience as more pleasant. The VAS was 15 cm wide, with labels reading "I don't feel pleasantness at all" on the left end and "I feel the most pleasant feeling I can imagine" on the right end. Responses were quantified by normalizing the horizontal click position to a value between 0 and 100, where 0 corresponded to the far left ("I don't feel pleasantness at all") and 100 corresponded to the far right ("I feel the most pleasant feeling I can imagine").

The experiment consisted of a familiarization phase followed by a test phase. During the familiarization phase, participants experienced five randomly selected conditions. In the test phase, there were two blocks, with each block consisting of 45 trials corresponding to the 45 distinct conditions. This resulted in a total of 90 trials. The order of the test conditions in a block was pseudo-randomized for each participant.

2.3 Result

We averaged the pleasantness rating scores for each condition (as shown in Figure 2). Due to the limited number of participants, we did not perform any statistical analyses. Preliminary results suggested that the influence of pointer speeds depended on the level of stiffness. Specifically, under no-stiffness conditions (0.0 N/mm), pleasantness peaked at a moderately slow press speed (0.5). While slow speeds generally evoke pleasant visually-induced resistance [8, 9], the lower rating at the slowest speed (0.25) implies that excessive slowing may cause frustration. This suggests that identifying an appropriate speed range is crucial, rather than simply minimizing speed.

Under the medium and high stiffness conditions (2.0 or 4.0 N/mm), the consistent effect of press speed was not observed. This suggests that the effect of illusory resistance was relatively small compared to physical stiffness. In contrast, pleasantness ratings tended to be higher at moderately high release speed ratios (e.g., 2.0) under high stiffness condition (4.0 N/mm). In future work, we

plan to conduct a full-scale experiment and perform detailed analyses.

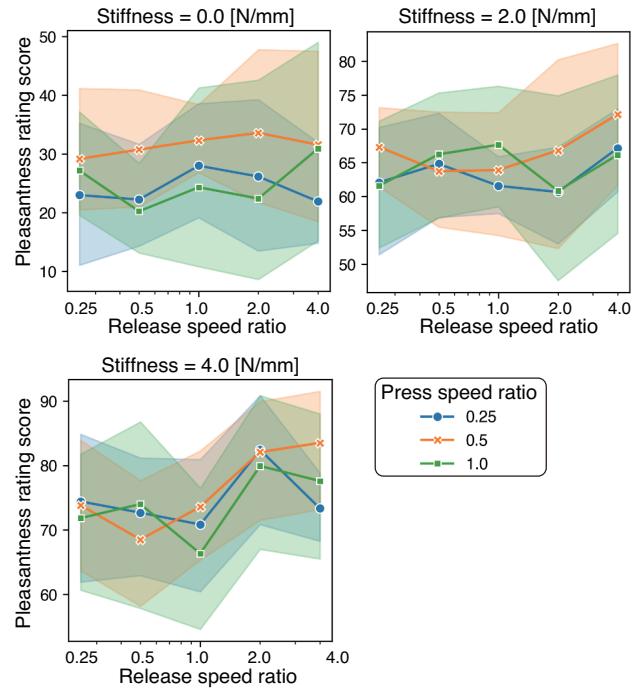


Figure 2: Effects of visual pointer speed on pleasantness under different stiffness conditions. Error bars denote 95% CI (N = 6).

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¹Obi Softbody 7 is a physics library for Unity developed by Virtual Method. It employs a position-based dynamics solver to simulate complex elastic and plastic soft bodies in real time. For details, see <https://assetstore.unity.com/packages/tools/physics/obi-softbody-130029>.